

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE						
Regular/Supplementary Winter Examination – 2024						
Course: B. Tech.		Branch: Chemical Engineering		Semester : 3		
Subject Code & Name: BTCHE305D- Renewable Energy Sources						
Max Marks: 60		Date: 21/02/2025		Duration: 3 Hr.		
Instructions to the Students: 1. Each question carries 12 marks. 2. Question No. 1 will be compulsory and include objective-type questions. 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6. 4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 5. Use of non-programmable scientific calculators is allowed. 6. Assume suitable data wherever necessary and mention it clearly.						
					(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)					12
1	What is renewable energy?				Remember	1
	a) Energy derived solely from fossil fuels	b) Energy that can be replenished naturally	c) Energy obtained from non replenishable sources	d) Energy that cannot be utilized more than once	/CO1	
2	Which of the following is not a renewable source of energy?				Remember	1
	a) Fossil fuels	b) Biomass	c) Sun	d) Wind	/CO1	
3	What is the primary/major useful component of biogas?				Remember	1
	a) Oxygen	b) Methane	c) Hydrogen	d) Carbon dioxide	/CO1	
4	What is anaerobic digestion?				Remember	1
	a) Produces biogas by heating the biomass	b) Produces biogas using micro-organisms operating in anaerobic conditions	c) Produces biogas by subjecting the biomass to high pressures	d) Produces biogas using micro-organisms operating in aerobic conditions	/CO1	
5	Which of the following temperature ranges are suitable for biomass gasification?				Remember	1
	a) Above 1000 degree Celsius	b) Between 500 and 600 degree Celsius	c) Between 700 and 1000 degree Celsius	d) Less than 500 degree Celsius	/CO2	
6	What is the principle behind tidal energy generation?				Remember	1

	a) Gravitational pull of the Moon and Sun	b) Photosynthesis in marine plants	c) Temperature differences in seawater	d) Wind currents over oceans	/CO3	
7	4. What is the basic working principle of direct solar dryers?				Applying	1
	a) Direct exposure of substance to fire	b) Direct use of sunlight to start a heater which is then used to dry the substance	c) Indirect exposure of substance to sunlight	d) Dehydrating the substance by directly exposing to sunlight	/CO2	
8	An instrument used to measure total solar radiation is called				Remember	1
	a) hygrometer	b) pyranometer	c) anemometer	d) speedometer	/CO2	
9	Which of the following is a commonly used material in solar cell				Remember	1
	a) aluminum	b) silicone	c) iron	d) copper	/CO2	
10	What is a fuel cell?				Applying	1
	a) Converts heat energy to chemical energy	b) Converts heat energy to electrical energy	c) Converts chemical energy to electrical energy	d) Converts kinetic energy to heat energy	/CO4	
11	What is the temperature difference used in ocean thermal energy conversion?				Applying	1
	a) 0 °C	b) 5 °C	c) 20 °C	d) 100 °C	/CO4	
12	How does electrolysis produce hydrogen?				Applying	1
	a) By running electricity to combine hydrogen and water	b) By separating water into hydrogen and oxygen and generating electricity	c) By passing electricity into water to separate it into hydrogen and oxygen	d) By passing electricity into water to evaporate it into hydrogen	/CO3	
Q. 2 Solve the following.						12
A)	What are the different types of renewable energy are practiced today? Which renewable energy source do you think is the best and why?				Applying	6
					/CO1	
B)	Explain the principle and working of a biogas plant. Draw sketch of the plant. What is the composition of the biogas?				Understan	6
					ding/CO1	
Q.3 Solve the following.						12
A)	What is solar cell? Explain construction and working of solar photovoltaic power plant.				Analyze	6
					/CO2	

B)	What are different solar collectors? Explain their working principle and advantages?	<i>Evaluate</i> /CO2	6
Q. 4	Solve Any Two of the following.		12
A)	What is wind farm? Explain construction and working of wind energy power plant. How much electricity does a wind turbine generate?	<i>Evaluate</i> /CO3	6
B)	Explain in details different types of wind turbines based on its position. Draw sketches of the same.	<i>Understan</i> <i>ding</i> /CO3	6
C)	What are the advantages, disadvantages and applications of wind energy?	<i>Understan</i> <i>ding</i> /CO3	6
Q.5	Solve Any Two of the following.		12
A)	Explain different components of Ocean Thermal Electric Conversion (OTEC) system to produce electricity.	<i>Understan</i> <i>ding</i> /CO3	6
B)	How energy is harvested from waves and tides of ocean?	<i>Understan</i> <i>ding</i> /CO3	6
C)	What are the advantages and challenges to produce electricity from ocean energy?	<i>Evaluate</i> /CO3	6
Q. 6	Solve Any Two of the following.		12
A)	What are the different hydrogen production methods? Explain any two of them in detail.	<i>Analyze</i> /CO4	6
B)	What are the different techniques that used to store hydrogen? What are the challenges for hydrogen storage and transportation?	<i>Analyze</i> /CO4	6
C)	Explain the construction and working of hydrogen fuel cell in detail?	<i>Understan</i> <i>ding</i> /CO4	6
	*** End ***		